

ALGEBRAIC MULTIGRID THEORY:
THE SYMMETRIC CASE

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by

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1. INTRODUCTION

The purpose of this work is to develop the theoretical background needed for *Algebraic MultiGrid* (AMG) solvers (see [B4, §13.1], [B6],[R1]) and at the same time to develop precise *algebraic theory* also for usual (geometric) multigrid solvers. The theory is more precisely quantitative than other rigorous approaches. Its insights are particularly important for cases where local mode analysis cannot be applied, such as AMG or geometric multigrid for problems with strongly discontinuous coefficients. The inability of mode analysis to give full insight to the multi-level solutions of such problems, and the pressing need for solving many such problems with fully efficient algorithms, that can only be guided by precise quantitative insights (realistically dependent on the various algorithmic parameters), were indeed the main motivations for this work.

The term "algebraic" multigrid, as against "geometric" multigrid, refers mainly to the mode of constructing coarser levels (coarser grids) in multi-level solvers. In the usual, *geometric* approach, each coarser grid is a simple, well-ordered coarsening of the next finer grid; e.g., the lines of the former are obtained by taking every other line of the latter, whether these are lines of gridpoints or lines of grid cells (when discretization is made in terms of cells). Sometimes only lines in some directions are so coarsened, while in other directions all fine-grid lines are retained in the coarser level [B4, §4.2.1]. In any case, the coarse-level variables, and their relation to the fine-grid variables (through interpolation), are characterized by their geometrical location.

In the *algebraic* approach no organized grids are assumed. The coarse-level variables are just a set of variables selected so as to satisfy certain criteria based on the fine-level equations. The most basic criterion is typically that each fine-level variable should be "strongly connected"

to some coarse-level variables (see Secs. 3.5, 4.3). The coarse-to-fine interpolations and the fine-to-coarse transfers are also mainly based on the algebraic connections (although "hybrid" schemes are envisioned in which some geometrical information is taken into account; see Sec. 5); in this respect AMG continues the line of development set in [A1] and [D1].

Each of these approaches has its advantages and domain of applicability. The geometric multigrid in many cases is more efficient in terms of storage and time, because it need not store the (sparse) matrix entries (both on fine and on coarser grid) and it need not assemble the coarse grid equations. But in other cases this storage and work are still invested even though the geometric algorithm is used. For such cases and some others, AMG has several advantages:

First, AMG can be used as a black-box solver. Since the algorithm is based on the given matrix, assuming only some general properties, it can be used for a wide range of problems, with no need to give special attention to boundary conditions, anisotropies, strong discontinuities in coefficients, etc., features which normally require advanced expertise. Moreover, since the AMG selection of the coarse grid is based on strong algebraic connections, extremely anisotropic equations, equations with strong discontinuities, quasi-elliptic equations (unsolvable by the usual algorithms; see [B5, §3.4]), and many other cases where full h -ellipticity is irregularly lost, are all solved by the same AMG algorithm, employing simple (pointwise) Gauss-Seidel relaxation. This is particularly important for problems where several of these difficulties appear together. In some pathological cases (see [A1, §8]) even 5-point diffusion equations cannot be solved with the usual, geometric coarsening.

An obvious use of AMG is for problems where no organized grids are employed and hence the geometric multigrid is not even applicable. This

includes finite elements discretization with arbitrary, irregular fine-grid triangulations, and large matrix equations which are not at all derived from continuous problems, such as the geodetic adjustment problem [M2], multivariate interpolation of scattered data [M1], large scale problems in economics (cf. references in [B6]), circuitry queuing problems, etc.

There may well be large-scale problems which cannot efficiently be solved by multi-level techniques. One of the purposes of [B6], [R1] and the present study is to define classes of problems for which AMG makes sense. In particular we study the role of the *geometrical origin* of the problem. It turns out that geometric background is important in order to guarantee coarse-levels sparsity; without it the number of non-zero entries per row in the coarse-level matrices may grow indefinitely (as function of the problem size). This role of the geometric background can be played implicitly, though; the algorithm need not use the geometrical information. This information is important for obtaining much better interpolations between levels (see Sec. 4.6), but there are again ways to use it implicitly (Sec. 5).

Despite the irregularity of the treated problems, the present theory is very realistic in its estimates, almost precise in studying the dependence of convergence rates on various factors. This is obtained by two technical devices. First, we introduce three *square error norms* $W = \sum a_{ii} e_i^2$, $E = \sum e_i r_i$ and $R = \sum r_i^2 / a_{ii}$, where e_i are the errors, r_i the residuals and a_{ii} the matrix diagonal terms, and show their precise interplay in the various multigrid processes. This interplay is particularly simple in the symmetric case: The coarse grid correction always reduces E and produces $R = CE$, with $C = O(1)$ if the interpolation satisfies a certain algebraic requirement. Relaxation always reduces E by γR , where $\gamma = O(1)$ and depends only on the ratio of off-diagonal sums to the diagonal terms in the

given matrix (or in an associated matrix, depending what relaxation is employed). This interplay holds even for general matrices; no regular grids are even needed.

When relaxation slows down we must therefore have $R \ll E$. This relation can be interpreted as the algebraic sense of the error "smoothness" (see Sec. 3.5).

In case the problem is sufficiently regular, better convergence rates can be obtained, because, if interpolation has a high enough order, the coarse-grid correction will reduce E to the order of the previous R ($\ll$ previous E).

The second technical device we use is to show that all the above interplay between E and R is basically *local*. This is particularly important for problems which are mostly regular, but feature some particular points or lines of singularities (reentrant corners, discontinuities in coefficients, etc.). The local nature of the relations between E and R entails that with some (negligible amount of) additional computations in the neighborhoods of singularities, the same convergence rates can be obtained as in the corresponding regular cases. This provides a rigorous justification for *local mode analysis*, through which the convergence rates of mostly-regular problems can precisely be predicted. It also provides a rigorous justification to a local *non-mode* analysis, based on the local interplay of E , R and W . Such an analysis is required to understand geometrical multigrid processes for equations with discontinuous coefficients, for which local *mode* analysis is sometimes useless.

2. CLASSES OF MATRICES

We deal with the solution of the linear system

$$\sum_{j=1}^n a_{ij}x_j = b_i, \quad (i=1, \dots, n) \quad (2.1)$$

or, in matrix notation

$$AX = b.$$

The matrix A can be fairly general, but for some of the processes described below to make sense we will see later that some properties of A are needed which are usually satisfied whenever the system of equations has a geometrical background; it may, but need not, arise from discretizing a continuous problem.

2.1 Local positive definiteness

In the present work we do not attempt to derive the most general theory. To make our study easier we start with the case that A is *symmetric*: $a_{ij}=a_{ji}$ (or, in the complex case, Hermitian: $a_{ij}=\overline{a_{ji}}$. Only in Theorem 3.4 this assumption is dropped.) Also, we assume A to be *positive definite* (in case of simple point relaxation. More general cases may be considered by using distributive and weighted relaxation schemes, in which case it is $\tilde{A}$, defined in Sec. 3.3, that should be symmetric and positive definite).

We will use, more specifically, the assumption that A is "*locally*" (or "*sparsely*") positive definite, i.e., for any vector x

$$x^T Ax = \sum_k f_k(x)^2, \quad (2.2)$$

where each $f_k(x)$ is a linear function involving only few of the components x_j . This class naturally involves least square problems with local (sparse) measurements. In addition we note in the next few sections several important examples of local positive definiteness.

2.2 Positive type matrices

The simplest case of local positive definiteness is that of positive-type matrices, i.e., matrices A for which

$$a_{ij} \leq 0 \text{ for all } j \neq i, \text{ and } \sum_j a_{jk} \geq 0 \text{ for all } i. \quad (2.3)$$

It is easy to see that

$$x^T Ax = \frac{1}{2} \sum_{i,j} (-a_{ij}) (x_i - x_j)^2 + \sum_i \left(\sum_j a_{ij} \right) x_i^2, \quad (2.4)$$

which is of the form (2.2).

2.3 Zero row sums

Sometimes, without loss of generality, it is convenient to assume that A has zero row sums, i.e.,

$$\sum_j a_{ij} = 0 \text{ for all } i. \quad (2.5)$$

Indeed, if (2.5) is not satisfied, we can always add a *slack variable*, x_0 say, taking

$$a_{0i} = a_{i0} = - \sum_{j=1}^n a_{ij}, \quad (i = 0, 1, \dots, n).$$

The new system clearly satisfies both (2.3) and (2.5). Note that with (2.5) the system is singular; the solution is determined only up to an additive constant. If x is a solution, so is also $x - (c, c, \dots, c)^T$. Thus, by choosing $c = x_0$, we can always obtain a solution in which $x_0 = 0$ and hence the non-slack variables satisfy the original system. Note also that, with the zero row sum assumption (2.5), relation (2.4) simplifies to

$$x^T Ax = \frac{1}{2} \sum_{i,j} (-a_{ij}) (x_i - x_j)^2. \quad (2.6)$$

The introduction of a slack variable is not just a theoretical artifact. It turns out to be a useful algorithmic device as well. Including the slack variable and the corresponding equation in the Gauss-Seidel relaxation is equivalent, in terms of the original system, to a step of adding the same constant to all unknowns so as to satisfy the sum of all the equations. Such a step is indeed needed in some cases (e.g., Helmholtz equation $-\Delta u + Cu = f$ with small $|C|$ and Neumann boundary conditions) in order to get fast convergence on the coarsest grid. On the finest grids this is usually not needed, so some work could be saved by avoiding there the introduction of the slack variable.

2.4 Block positive type matrices

A more general class of matrices which are locally positive definite are the *block* positive type matrices. Partitioning A into (small) square blocks A_{IJ} , we say that A is block positive type if $A_{IJ} + A_{JI}$ is (possibly non-symmetric) negative semi-definite for all $I \neq J$, while $\sum_J (A_{IJ} + A_{JI})$ is positive semi-definite for all I .

In addition to the positive-type matrices (2.3), the new class includes some interesting cases, such as the matrix of the x-y geodetic problem [B6, § 6.5]. The stably discretized steady-state Navier-Stokes equations, whether compressible or incompressible, are also block positive type, when properly signed, although the system is symmetric only for special cases (Stokes equations).

For a block positive type matrix A , let $\hat{x}_I$ denote the corresponding I -th block of an arbitrary vector x . Then

$$\begin{aligned}
x^T A x &= \sum_{I,J} \hat{x}_I^T A_{IJ} \hat{x}_J \\
&= \frac{1}{4} \sum_{I,J} (\hat{x}_I - \hat{x}_J)^T (-A_{IJ} - A_{JI}) (\hat{x}_I - \hat{x}_J) \\
&\quad + \frac{1}{2} \sum_I \hat{x}_I^T \sum_J (A_{IJ} + A_{JI}) \hat{x}_I,
\end{aligned} \tag{2.7}$$

which clearly has the required form (2.2).

2.5 Essentially positive type matrices

Quite often the matrix A is "almost" positive type, in the sense that $\sum_j a_{ij} \geq 0$ is still true for all i , but the requirement $a_{ij} \leq 0$ is violated for some $i \neq j$. If those positive a_{ij} are sufficiently small, they need not bother us. This, for example, is usually the situation in discretizing second-order elliptic equations with mixed terms ($\partial^2 u / \partial x_1 \partial x_2$ etc.). For each $i \neq j$ with $a_{ij} > 0$, we can then usually find another variable, k say, such that both $-a_{ik} > \alpha a_{ij}$ and $-a_{jk} > \beta a_{ij}$, where $\alpha > 1$, $\beta > 1$ and $(\alpha-1)(\beta-1)=1$. We can then replace $(x_i - x_j)^2$ in (2.4) by

$$\alpha(x_i - x_k)^2 + \beta(x_j - x_k)^2 - (\bar{\alpha}x_i + \bar{\beta}x_j - (\bar{\alpha} + \bar{\beta})x_k)^2 \tag{2.8}$$

where $\bar{\alpha} = (\alpha-1)^{1/2}$ and $\bar{\beta} = (\beta-1)^{1/2}$, so that the resulting quadratic will have the form (2.2). Whenever we can so repair (2.4) to get the form (2.2), we say that A is *essentially positive type*.

Analogously, if (2.7) can similarly be repaired to have the form (2.2), we say that A is *essentially block positive type*.

2.6 RB matrices and RB vectors

Another important class of matrices, which are used below mainly as examples for the sharpness of some estimates, is the class of RB (Red-Black)

matrices. These are defined as matrices A for which there exist two disjoint subsets S_R and S_B such that $S_R \cup S_B = \{1, 2, \dots, n\}$ and such that $a_{ii}a_{ij} \geq 0$ if i and j are in the same subset and $a_{ii}a_{ij} \leq 0$ if i and j are in different subsets.

All positive-type matrices with Property A (see [Y1]), for example, are RB matrices, including all those arising from the common $(2d+1)$ -point discretization of second-order elliptic equations in d -dimensional space. In addition, the common discretization of the biharmonic operator, the staggered discretization of the compressible or incompressible Navier-Stokes equations (the principal and subprincipal terms; see [B3]), and perhaps most other discretized problems, yield RB matrices, not necessarily of positive type.

For a given RB matrix, a vector x will be called an RB *vector* if $x_i = -x_j$ whenever $i \in S_R$ and $j \in S_B$.

3. RELAXATION AND SMOOTHING

3.1 Gauss-Seidel relaxation

Let x be some approximation to the solution X of (2.1). We denote the error by $e = x - X$ and the residual by $r = Ae$. Thus, $r_i = \sum_j a_{ij} x_j - b_i$. We will use the square error norms

$$E = e^T r \quad \text{and} \quad R = \sum_i |r_i|^2 / a_{ii}$$

and observe their changes in various processes. (The superscript T stands for transposed, and in case of complex values, for complex-conjugate transposed.)

Consider a Gauss-Seidel relaxation sweep. Let e^0 , r^0 , E^0 and R^0 denote the values of e , r , E and R , respectively, before the sweep, and e^1 , r^1 , E^1 and R^1 the corresponding values after the sweep. And let r_j^* denote the "dynamic residual," i.e., the value of r_j just before the step of relaxing x_j , and denote $R^* = \sum_j |r_j^*|^2 / a_{jj}$. Here, "relaxing x_j "

means the step of replacing x_j by a new value $x_j + \delta_j$, as a result of which each r_i is replaced by $r_i + a_{ij}\delta_j$, ($i=1, \dots, n$). Taking

$$\delta_j = -r_j^*/a_{jj} \quad (3.1)$$

we see that, in particular, r_j is replaced by 0. Using symmetry (this is in fact the main point where symmetry is used), it is also easy to see that E is thereupon replaced by

$$E + \sum_i \bar{e}_i a_{ij} \delta_j = E + \bar{r}_j \delta_j = E - \frac{|r_j^*|^2}{a_{jj}}. \quad (3.2)$$

(The complex-conjugate bars are shown in case one is interested in complex functions, like those appearing in the corresponding mode analysis.) The Gauss-Seidel sweep consists of relaxing x_1 , then x_2 , then x_3 , etc. up to x_n . Hence,

$$E^1 = E^0 - R^*. \quad (3.3)$$

Thus, in a full sweep the value of E is decreased by a quantity R^* , akin to R^0 and to R^1 . Actually, we now prove that the decrease R^* is comparable to both R^0 and R^1 .

THEOREM 3.1. If A is symmetric (or Hermitian) then $E^1 \leq E^0 - \gamma_0 R^0$, where $\gamma_0 = [(1+\gamma_-)(1+\gamma_+)]^{-1}$ and

$$\gamma_- = \gamma_-(A) = \max_i \sum_{j < i} |a_{ij}|/a_{ii}, \quad \gamma_+ = \gamma_+(A) = \max_i \sum_{j > i} |a_{ij}|/a_{ii}. \quad (3.4)$$

Proof. After relaxing x_i we have a zero residual at i ; that is

$$r_i = r_i^0 + \sum_{j \leq i} a_{ij} \delta_j = 0. \quad (3.5)$$

Hence, by the Cauchy-Schwarz inequality and (3.4),

$$\begin{aligned}
\sum_i |r_i^0|^2 / a_{ii} &= \sum_i a_{ii}^{-1} \left| \sum_{j \leq i} a_{ij} \delta_j \right|^2 \\
&\leq \sum_i a_{ii}^{-1} \left(\sum_{j \leq i} |a_{ij}| \right) \left(\sum_{j \leq i} |a_{ij}| |\delta_j|^2 \right) \\
&\leq (1+\gamma_-) \sum_i \sum_{j \leq i} |a_{ij}| |\delta_j|^2 \\
&= (1+\gamma_-) \sum_j |\delta_j|^2 \sum_{i \geq j} |a_{ij}| \\
&\leq (1+\gamma_-) (1+\gamma_+) \sum_j a_{jj} |\delta_j|^2 \\
&= (1+\gamma_-) (1+\gamma_+) R^*
\end{aligned}$$

(using (3.1)), and the theorem follows from (3.3).

COROLLARY 3.1. *If A is symmetric and positive definite then*

$$E \geq \gamma_0 R.$$

This follows from Theorem 3.1, since $E^1 \geq 0$. Numerical values of γ_0 and the sharpness of the estimate are discussed in the next Section. The following is a complementary theorem which provides an important additional insight.

THEOREM 3.2. *If A is symmetric (or Hermitian) then $E^1 \leq E^0 - \gamma_1 R^1$, where $\gamma_1 = [\gamma_-(A) \gamma_+(A)]^{-1}$.*

Proof. After relaxing x_i we have $r_i = 0$, hence

$$r_i^1 = \sum_{j > i} a_{ij} \delta_j. \quad (3.6)$$

By Cauchy-Schwarz inequality

$$\begin{aligned}
 \sum_i \left| r_i^1 \right|^2 / a_{ii} &= \sum_i a_{ii}^{-1} \left| \sum_{j>i} a_{ij} \delta_j \right|^2 \\
 &\leq \sum_i a_{ii}^{-1} \left(\sum_{j>i} |a_{ij}| \right) \left(\sum_{j>i} |a_{ij}| \delta_j^2 \right) \\
 &\leq \gamma_+ \sum_i \sum_{j>i} |a_{ij}| \delta_j^2 \\
 &\leq \gamma_+ \gamma_- R^*
 \end{aligned}$$

and the theorem follows from (3.3).

3.2 Sharpness of bounds and converse theorem

The constants γ_0 and γ_1 in Theorems 3.1 and 3.2 are practically optimal. In the most important cases the stated bounds are actually attained, and in other cases they are reasonably approached. Moreover, in any case, constants as close to practical as one wishes can be derived. (The reader not interested in these details can skip this section.)

Consider first the case of an RB matrix A (defined in Sec. 2.6), where

$$\gamma_- = \sum_{j<i} |a_{ij}|/a_{ii}, \quad \gamma_+ = \sum_{j>i} |a_{ij}|/a_{ii} \quad (3.7)$$

for all i . Then, for an RB vector of corrections δ , the inequalities in the above proofs all become equalities. Hence, γ_0 and γ_1 are the best possible constants in terms of γ_- and γ_+ . Moreover, for an RB matrix A and an RB vector δ , if (3.7) is *approximately* satisfied, in the sense that for some small subset $S_\delta \subset \{1, 2, \dots, n\}$ and for a small positive constant ϵ we have

$$\sum_{j<i} |a_{ij}|/a_{ii} \geq \gamma_- - \epsilon, \quad \sum_{j>i} |a_{ij}|/a_{ii} \geq \gamma_+ - \epsilon \quad (3.8)$$

for all $i \notin S_\delta$, then the inequality in the proofs are not far from equalities, hence the bounds γ_0 and γ_1 are almost attained. This is indeed the usual

situation in geometrically ordered relaxation of finite difference approximations to differential equations on regular grids with meshsize h , where S_0 is a subset of points near the boundary and $\epsilon \rightarrow 0$ as $h \rightarrow 0$ if the coefficients of the differential equation are continuous. The bounds γ_0 and γ_1 are then attained in the limit $h \rightarrow 0$.

The above class of matrices include most of the regular symmetric matrices arising in differencing differential equations on regular grids. For matrices not in that class the bounds are not generally approached, but the practical values of γ_0 and γ_1 are not far from the stated ones, provided the definition of γ_+ and γ_- are slightly modified by excluding some special values of i from the maximization in (3.4).

Moreover, for regular and large ($h \rightarrow 0$) grids, precise values of γ_0 and γ_1 can always be derived by local mode analysis (cf. [B4]). Usually, values calculated by local mode analysis are rigorously true only for the infinite domain (or the rectangular, etc.) problem. In determining γ_0 and γ_1 , however, one can rigorously justify the mode analysis values, $\tilde{\gamma}_0$ and $\tilde{\gamma}_1$ say, for any domain, at least in the limit of vanishing meshsize ($h \rightarrow 0$) and with proper modification of the relaxation scheme at a small number of points. This is done in the following way: Given a small positive constant ϵ , before each relaxation sweep a (possibly large) number $\nu(\epsilon)$ of relaxation sweeps is performed over all those gridpoints whose distance from the boundary is less than $k(\epsilon)h$. With a proper selection of $\nu(\epsilon)$, $k(\epsilon)$ and $h_0(\epsilon)$, we can prove that for any $0 < h \leq h_0(\epsilon)$ Theorems 3.1 and 3.2 are satisfied with $\gamma_0 = \tilde{\gamma}_0 - \epsilon$ and $\gamma_1 = \tilde{\gamma}_1 - \epsilon$. The detailed proof is omitted since it is not used below.

For symmetric constant-coefficients difference operator L^h with *geometrically consistent relaxation ordering* (i.e., if grid point $\underline{x}$ is relaxed after $\underline{x}-\underline{a}$, then it is relaxed before $\underline{x}+\underline{a}$), local mode analysis

yields particularly simple results: $L^h \exp(i\theta x/h)$ can be written as $[a + b(\theta) + \bar{b}(\theta)] \exp(i\theta x/h)$, where $a = a_{ij}$, and each relaxation sweep transforms the error component $\exp(i\theta x/h)$ to $(a+b)^{-1} \bar{b} \exp(i\theta x/h)$. Hence, it is easy to calculate that for each error component

$$\frac{E^0 - E^1}{R^0} = \left| \frac{a}{a+b} \right|^2, \quad \frac{E^0 - E^1}{R^1} = \left| \frac{a}{b} \right|^2. \quad (3.9)$$

Since $b(0) = \bar{b}(0) = -a/2$, it follows that for smooth components ($\theta \neq 0$) we have $E^0 - E^1 \approx 4R^0 \approx 4R^1$. Indeed, when relaxation is slow it is clear that $r^0 \approx r^1 \approx r^*/2$.

There are, of course, particular error components for which $E^0 - E^1$ is considerably larger than $\gamma_0 R^0$, and (other) components for which $E^0 - E^1$ is much larger than $\gamma_1 R^1$. Still, the lower bounds of Theorems 3.1 and 3.2 are very realistic, because they are (almost) attained exactly for the components which are normally dominant at the stage for which each theorem is used. Theorem 3.1 is used to estimate the drop in E at the first relaxation sweep after the coarse-grid correction. Due to the nature of interpolation, the dominant errors e^0 at this stage are normally highly oscillatory (e.g., with a large RB component) which are indeed the type of errors for which the bound in Theorem 3.1 is attained or approached. Theorem 3.2, on the other hand, is used to estimate the smallness of R when relaxation slows down. At this stage the dominant errors are smooth (see Sec. 3.5), for which the bound of Theorem 3.2 is indeed attained (e.g., when A is positive-type), or reasonably approached.

For continuous symmetric positive type operators, with geometrically consistent relaxation ordering, it is easy to see that $\gamma_-(A) \approx \gamma_+(A) \approx 1/2$. Hence, $\gamma_0 \approx 4/9$ and $\gamma_1 \approx 4$, values which are actually attained for the

highest- and lowest-frequency errors, respectively. For this type of operators, as well as many others, the following theorem, a converse of Theorem 3.1, can be of interest.

THEOREM 3.3. If A is symmetric and $\gamma_-(A)\gamma_+(A) < 1$, then $E^0 - E^1 \leq \tilde{\gamma}_0 R^0$, where $\tilde{\gamma}_0 = [1 - (\gamma_- \gamma_+)^{1/2}]^{-2}$.

Proof. Applying Cauchy-Schwarz inequality to

$$r_i^* = r_i^0 + \sum_{j < i} a_{ij} \delta_j$$

and then using (3.1), it follows that

$$R^* \leq R^0 + \frac{\gamma_+ \gamma_-}{4} R^* + 2(\gamma_- \gamma_+ R^0 R^*)^{1/2},$$

yielding, by (3.3), the theorem.

In particular, for symmetric positive type operators with constant coefficients and with consistent relaxation ordering we have $\gamma_+ = \gamma_- = 1/2$, hence we get the interesting result

$$4R^1 \leq E^0 - E^1 \leq 4R^0. \quad (3.10)$$

When relaxation slows down then $R^1 \approx R^0$, hence we have quite a precise measure for the relation between that slowness and the smallness of ratio R/E . Namely, the convergence factor is almost precisely $1 - 4R/E$. As we saw above (following (3.9)), this is in practice the slow convergence factor even when the operator is not of positive type.

3.3 Other point relaxation schemes

Theorems 3.1 and 3.2 can be extended to perhaps any reasonable point relaxation, including point Jacobi (Richardson) schemes, red-black Jacobi (even with couplings within each color), etc. In fact, if sufficiently small,

but depending only on

$$\gamma_*(A) = \max_i \sum_j |a_{ij}|/a_{ii}, \quad (3.11)$$

under-relaxation parameter ω is used (as in most multigrid theories, but not in practice), then the proof is trivial, since R^0 , R^* and R^1 are obviously comparable, and the effect of relaxing points simultaneously is $O(\omega\gamma_*(A))$ compared to relaxing them successively.

For point Jacobi, for example, from $\delta_i = -\omega r_i^0 / a_{ii}$ we get

$$E^1 = E^0 - 2\omega \sum_i |r_i^0|^2 / a_{ii} + \omega^2 \sum_{i,j} \frac{a_{ij}}{a_{ii}a_{jj}} r_i^0 r_j^0$$

hence, using Cauchy-Schwarz inequality,

$$E^1 \leq E^0 - \omega[2 - \omega\gamma_*(A)] R^0. \quad (3.12)$$

Note that for positive type operators $\gamma_*(A) = 2$, hence in this case $\gamma_0 = 2\omega(1-\omega)$. This value of γ_0 is again sharp; it is attained for any RB matrix A and RB error e (see Sec. 2.6).

Other types of point relaxation are the *distributive schemes* (see [B5, §5.3], [B2, §4.1] or [B4, §3.4]) and *weighted schemes* (see [B1, §3.3] or [B4, §3.4]), and the combination of distributive and weighted relaxation. Such schemes should be used whenever $\gamma_*(A)$ is too large. Each such scheme is equivalent to a simple (non-distributive and not weighted) scheme for another matrix $\tilde{A} = BAC$, hence its analysis follows from applying the above theory to $\tilde{A}$ (if $\tilde{A}$ is symmetric). The distributed Gauss-Seidel (DGS) scheme, for example, is equivalent to a Gauss-Seidel relaxation of $\tilde{A} = AC$ (see [B2, §4.1]). *Kaczmarz relaxation* [T1], in particular, is equivalent to Gauss-Seidel relaxation for $\tilde{A} = AA^T$. Thus, applying Theorem 3.1 to $\tilde{A} = AA^T$, one immediately deduces

THEOREM 3.4. For any matrix A (not necessarily symmetric), let

$$\tilde{E} = \sum_i |e_i|^2 \quad \text{and} \quad \tilde{R} = \sum_i (|r_i|^2 / \sum_\lambda |a_{i\lambda}|^2) \quad (3.13)$$

and let $\tilde{E}^0$ and $\tilde{R}^0$ be the initial values of these square norms and $\tilde{E}^1$ and $\tilde{R}^1$ their values after one Kaczmarz relaxation sweep. Then

$$\tilde{E}^1 \leq \tilde{E}^0 - \tilde{\gamma}_0 \tilde{R}^0 \quad (3.14)$$

and also

$$\tilde{E}^1 \leq \tilde{E}^0 - \tilde{\gamma}_1 \tilde{R}^1 \quad (3.15)$$

where

$$\tilde{\gamma}_0 = [(1 + \tilde{\gamma}_-) (1 + \tilde{\gamma}_+)]^{-1}$$

$$\tilde{\gamma}_1 = (\tilde{\gamma}_- \tilde{\gamma}_+)^{-1}$$

$$\tilde{\gamma}_- = \max_i \left(\sum_{j < i} \left| \sum_\lambda a_{i\lambda} \bar{a}_{j\lambda} \right| \right) / \sum_\lambda |a_{i\lambda}|^2$$

$$\tilde{\gamma}_+ = \max_i \left(\sum_{j > i} \left| \sum_\lambda a_{i\lambda} \bar{a}_{j\lambda} \right| \right) / \sum_\lambda |a_{i\lambda}|^2$$

Observe that $\tilde{\gamma}_\pm$ are reasonably small in many cases where $\gamma_\pm$ of Theorem 3.1 are not. In fact, for any reasonable discretization of differential equations $\tilde{\gamma}_+$ and $\tilde{\gamma}_-$ are smaller than 2 or so.

3.4 Block relaxation

In case of block relaxation, such as line relaxation or collective relaxation, the theory above still applies, but in block notation: Denoting by E_i the i -th block of e , by A_{ij} the corresponding blocks of A , a Gauss-Seidel step for the i -th block replaces E_i by $E_i - A_{ii}^{-1} R_i^*$, where $R_i = \sum_j A_{ij} E_j$ and R_i^* is the value of R_i just *before* this step. Hence, analogously to (3.2),

this step reduces the error energy E by

$$R_i^{*T} A_{ii}^{-1} R_i^* . \quad (3.16)$$

The full block Gauss-Seidel sweep (making such a step successively for all the blocks) will therefore reduce E by

$$\hat{R}^* = \sum_i R_i^{*T} A_{ii}^{-1} R_i^* . \quad (3.17)$$

From here one can prove, analogously to the proofs of Theorems 3.1 and 3.2, the following theorem.

THEOREM 3.5. Let $\hat{R}^0$ and $\hat{R}^1$ respectively be the values of the block residual norm

$$\hat{R} = \sum_i R_i^T A_{ii}^{-1} R_i$$

before and after a block-Gauss-Seidel relaxation sweep, for a symmetric matrix A . Then the decrease $E^0 - E^1$ of the error energy E due to that sweep is at least $\hat{\gamma}_0 \hat{R}^0$, and at least $\hat{\gamma}_1 \hat{R}^1$, where

$$\hat{\gamma}_0 = (\hat{\gamma}_+^! \hat{\gamma}_-^!)^{-1}, \quad \hat{\gamma}_1 = (\hat{\gamma}_+ \hat{\gamma}_-^!)^{-1}$$

$$\hat{\gamma}_- = \max_i (\zeta_i \sum_{j < i} \alpha_{ij}), \quad \hat{\gamma}_-^! = \max_i (\zeta_i \sum_{j \leq i} \alpha_{ij})$$

$$\hat{\gamma}_+ = \max_i (\zeta_i \sum_{j > i} \alpha_{ij}), \quad \hat{\gamma}_+^! = \max_i (\zeta_i \sum_{j \geq i} \alpha_{ij})$$

ζ_j being the spectral radius of A_{jj}^{-1} and α_{ij} being the L_2 norm of A_{ij} .

This theorem, too, can be extended to other block relaxation schemes, such as block-Jacobi, zebra and distributed block relaxation schemes, the only change being the form of dependence of $\hat{\gamma}_0$ and $\hat{\gamma}_1$ on $\hat{\gamma}_+$ and $\hat{\gamma}_+^!$. The derived constants are again sharp in various cases, while in other cases

they are not far off, and optimal constants can be derived from local mode analyses, rigorously justifiable for sufficiently large ($h \rightarrow 0$) regular grids and geometric ordering of relaxation.

Concerning simultaneous relaxation, we would like to add here another result, which is an interesting generalization of formula (3.2) above and of Corollary 3.1, and which will be useful in future work.

THEOREM 3.6. *If A is symmetric and positive definite, then relaxing simultaneously the I-th block reduces E at least by*

$$\gamma_0^I \sum_{i \in I} |r_i^*|^2 / a_{ii}$$

where

$$\gamma_0^I = [(1 + \gamma_-^I) (1 + \gamma_+^I)]^{-1}$$

$$\gamma_-^I = \max_{i \in I} \sum_{\substack{j < i \\ j \in I}} |a_{ij}| / a_{ii}$$

$$\gamma_+^I = \max_{i \in I} \sum_{\substack{j > i \\ j \in I}} |a_{ij}| / a_{ii}$$

The proof follows from (3.16) and from applying Corollary 3.1 to the matrix A_{II} (instead of A) and to the error $A_{II}^{-1} R_I^*$ (instead of e).

3.5 Algebraic sense of smoothing

A basic premise of multigrid processing is that relaxation is very efficient in smoothing the error, which makes it possible to efficiently approximate relaxed errors on coarser grids. More precisely, it is asserted that when the convergence of relaxation becomes slow, the error must be smooth in some sense. It is instructive to see that this smoothing has a well defined algebraic meaning. (This discussion is heuristic. It is not used in the subsequent multigrid theory, except as a motivation.)

Indeed, by Theorem 3.2, when the convergence of Gauss-Seidel relaxation slows down, i.e., when $E^0 - E^1 \ll E^0$, we must have

$$R \ll E. \quad (3.18)$$

This by itself can be regarded as an expression of smoothness of e , but we can give it even more intuitive interpretations. Consider for example the positive-type case (2.3). By the Cauchy-Schwarz inequality

$$E^2 \leq R \sum a_{ii} e_i^2, \quad (3.19)$$

hence, the slow convergence condition (3.18) implies

$$E \ll \sum a_{ii} e_i^2, \quad (3.20)$$

or, by (2.4),

$$\sum_{i,j} (-a_{ij}) (e_i - e_j)^2 \ll \sum a_{ii} e_i^2. \quad (3.21)$$

Since $a_{ij} \leq 0$ for $i \neq j$, this means that $e_i - e_j$ is on the average small compared with $\|e\|$ whenever $|a_{ij}|$ is comparable with either $\max_{k \neq i} |a_{ik}|$ or $\max_{i \neq j} |a_{jk}|$, i.e., whenever i and j are "*strongly coupled*". We can thus say that the Gauss-Seidel relaxation (and similarly the other schemes) *smoothes the error along the strongest couplings* (and couplings comparable to the strongest).

Stronger smoothing is of course obtained by block relaxation. When such relaxation slows down we must have, by Theorem 3.5,

$$\hat{R} \ll E. \quad (3.22)$$

Hence, applying now the Cauchy-Schwarz inequality to E in the form

$E = \sum (A_{\ell\ell}^{-1/2} E_\ell)^T (A_{\ell\ell}^{-1/2} R_\ell)$, we obtain

$$\sum_{i,j} (-a_{ij}) (e_i - e_j)^2 \ll \sum E_\ell^T A_{\ell\ell} E_\ell. \quad (3.23)$$

Hence, for $(e_i - e_j)$ to be small on the average, it is enough that a_{ij} is comparable to

$$\min (\max_{k \notin \ell_i} |a_{ik}|, \max_{k \notin \ell_j} |a_{jk}|),$$

where ℓ_i is the block containing i . Thus, *by including the strongest couplings in the blocks, block relaxation will smooth the error also along next-strongest couplings.*

We will indeed see later, more rigorously, that we obtain good multigrid rates if we see to it that our interpolation is along couplings which are about strongest (or next-strongest) in the above senses. From here follows the *rule of block relaxation* [B4, §3.3].

Although the smoothness may more easily be visualized through (3.21) or (3.23), it is the stronger conditions (3.18) or (3.22) which we will actually use in our theory below. In fact, from Example 4.2 below we can see that for algebraic multigrid the weaker conditions are not enough.

Condition (3.21) is indeed fundamentally stronger than (3.18), since it implies small changes in *derivatives* of e along strong couplings, while the latter implies only small changes in e itself.

For simplicity, we have discussed smoothing for positive type matrices only. The *generalization to locally positive definite matrices* (see Sec. 2.1) is straightforward. Instead of (2.6) one generally uses (2.2) in the left-hand side of (3.21), showing that when relaxation slows down $f_k(e)$ must on the average be small compared with $a_{ij} e_i^2$. In case of essentially

block positive type matrices (Secs. 2.4, 2.5) this still means smallness of certain error differences compared with the error itself (see (2.7)): Strong coupling here means eigenvectors with large eigenvalues of $-A_{IJ}-A_{JI}$ (compared with other eigenvalues of any $-A_{IK}-A_{KI}$ or $-A_{JK}-A_{KJ}$), and components of $\xi_I-\xi_J$ along those eigenvectors should on the average be small. More generally the smallness of $f_k(e)$ does not necessarily mean smallness of error differences, but can still be used for devising the coarse-to-fine interpolation.

In various cases of locally positive type matrices the linear forms f_k are not explicitly known. Practical ways of devising interpolation when f_k 's are not given are discussed in [R1].

4. TWO-LEVEL ALGEBRAIC ANALYSIS

4.1 General description and notation

We have thus seen that when convergence by relaxation is slow, the error e must be smooth, in the sense that along strong algebraic couplings it has small variations (or sometimes in a more general sense). We can therefore hope to get good approximations to that error "on a coarser grid," i.e., by solving a much smaller system of equations. For that purpose we have to construct that "coarse level" system of equations and specify the "interpolation," i.e., the relation by which an approximation to e (the "fine level" error) is obtained from the coarse level solution. In geometrical setting there are several quite natural ways for selecting the coarse level and the interpolation, but in purely algebraic terms the methods should be different.

We will generally denote the coarse-level vector of unknowns by $e^c = (e_1^c, e_2^c, \dots, e_n^c)^T$ and assume a coarse-to-fine linear interpolation I_c given by

$$(I_c e^c)_i = \sum_{k=1}^{n^c} w_{ik} e_k^c, \quad (1 \leq i \leq n). \quad (4.1)$$

Thus, $I_c e^c$ is a fine-level vector, intended to approximate the fine-level error vector e .

We can usually (but not necessarily) think of e^c as approximating a properly chosen subset of e , with each e_k^c approximating a particular $e_{F(k)}$. The smoothness properties of e resulting from relaxation, as described above, should be used in selecting this subset and in choosing the weights w_{ik} . In particular they imply that w_{ik} should be large only when i and $j=F(k)$ are *strongly connected*. By this we roughly mean that there exists a short chain $k_0, k_1, \dots, k_\ell$ (usually $\ell=1$ or 2) such that $k_0=i$, $k_\ell=j$ and k_α is strongly coupled with $k_{\alpha-1}$, ($\alpha=1, \dots, \ell$).

At this point we will not prescribe, however, any exact rules for choosing the weights. Instead, we will ask our theory below to furnish conditions that should generally be satisfied by the weights. Hence, the theory will apply to regular (geometric) multigrid as well.

Having computed e^c , its interpolant $I_c e^c$ is used to correct the fine-level approximation. In this section we will use superscripts $-$ and 0 , respectively, to denote the fine-level parameters before and after this correction. In particular, e^- , r^- , E^- and R^- will respectively denote the values of the error vector e , the residual vector r and the square norms E and R *before* the coarse-level correction, while e^0 , r^0 , E^0 and R^0 will denote the corresponding values *after* that correction. Hence $e^0 = e^- - I_c e^c$, $r^\ell = A e^\ell$, $E^\ell = \sum_i e_i^\ell r_i^\ell$ and $R^\ell = \sum_i |r_i^\ell|^2 / |a_{ii}|$, ($\ell = -, 0$).

What equations should be satisfied by e^c ? In regular (geometric) multigrid situations the cheapest way for deriving the coarse-level equations is usually to write coarse-grid approximations to the original differential equations (see more about this issue in [B4, §11]). In algebraic multigrid

this approach is not applicable, and a general prescription in the symmetric case (see [B6]) is to require e^C to satisfy

$$(I_C)^T A I_C e^C = (I_C)^T r^-. \quad (4.2)$$

This is equivalent to choosing that e^C for which E^0 will be minimal. It therefore implies the following three properties of the coarse-level correction:

- (i) $E^0 \leq E^-$.
- (ii) $E^0 \leq \tilde{E}^0$, where $\tilde{E}^0$ is the value of E after a correction by any other coarse-level vector $\tilde{e}^C$.
- (iii) e^0 is a projection of e^- ; i.e., the linear transformation $e^0 = P e^-$ satisfies $e^0 = P e^0$. Hence we must have $(I_C)^T r^0 = 0$, or, by (4.1),

$$\sum_j w_{jk} r_j^0 = 0 \quad \text{for all } k. \quad (4.3)$$

4.2 Relaxability after coarse level correction

In this section we show conditions under which the "energy" E^0 of the error e^0 produced after the coarse-level correction is efficiently reducible by subsequent relaxation sweeps. Since the coarse-level correction itself reduces E , this is equivalent to showing the efficiency of the two-level cycle (composed of a coarse-level correction followed by relaxation sweeps). By "efficiency" here we mean that E is reduced to $E \leq CE$, where $C < 1$ and C does not depend on the size of the matrix A . Actually, C will depend only on "local" constants like $\gamma_*(A)$ (see (3.11)). We thus say that e^0 is "*relaxable*" if a relaxation sweep will reduce E^0 to $E^1 \leq CE^0$ with such C . From Secs. 3.1 and 3.3 it follows that e^0 is "*point-relaxable*", i.e., relaxable by a point relaxation (Gauss-Seidel, under-relaxation Jacobi, etc.) if (and usually only

if--see Sec. 3.2) one can ensure that $R^0 \geq C_0 E^0$. This relation, and the size of its constant, are governed by the choice of w_{jk} , as the following theorem shows.

THEOREM 4.1. *If there is a constant C_0 such that for any fine level vector e there exists a coarse level vector e^c for which*

$$C_0 \sum_j a_{jj} (e_j - \sum_k w_{jk} e_k^c)^2 \leq e^T A e \quad (4.4)$$

then $R^0 \geq C_0 E^0$.

In the above summations j extends over the fine-level and k over the coarse level, of course. This convention is also kept in the summations below.

Proof: Choosing an e^c which satisfies (4.4) with $e=e^0$ and using (4.3), one obtains

$$\begin{aligned} E^0 &= \sum_j e_j^0 r_j^0 - \sum_k e_k^c \sum_j w_{jk} r_j^0 \\ &= \sum_j r_j^0 (e_j^0 - \sum_k w_{jk} e_k^c). \end{aligned}$$

Hence, by the Cauchy-Schwarz inequality and (4.4)

$$(E^0)^2 \leq R^0 \sum_j a_{jj} (e_j^0 - \sum_k w_{jk} e_k^c)^2 \leq R^0 E^0 / C_0$$

and hence $R^0 \geq C_0 E^0$.

COROLLARY 4.1. *If I_c is a regular averaging, i.e.,*

$$w_{jk} \geq 0 \quad \text{and} \quad \sum_k w_{jk} = 1, \quad (4.5)$$

and if for any e there exists e^c such that

$$C_0 \sum_{j,k} a_{jj} w_{jk} (e_j - e_k^c)^2 \leq e^T A e \quad (4.6)$$

then $R^0 \geq C_0 E^0$

This follows from Theorem (4.1) upon replacing e_j in (4.4) by $\sum_k w_{jk} e_j$ and applying Cauchy-Schwarz inequality.

4.3 Conclusions for positive type matrices

The theorem and the corollary spell out various possible rules of interpolation that can guarantee relaxability. For simplicity we first discuss such rules for the positive-type (2.3) zero-row-sum (2.5) case. In this case interpolation is chosen to be a usual averaging (4.5), and the required inequality (4.6) can be written, by (2.6), as

$$\sum_{j,i} (-a_{ji})(e_j - e_i)^2 \geq C_0 \sum_{j,k} a_{jj} w_{jk} (e_j - e_k^c)^2. \quad (4.7)$$

The zero row sum property (2.5) is, of course, not essential. Even without that property it is enough to satisfy (4.7), as is evident from (2.4). But the addition of the slack variable makes it *easier* to satisfy (4.7) for a *larger* C_0 . It makes it possible to exploit strong couplings to the slack.

A simple way to satisfy (4.7) is to think of e^c as corresponding to a subset of e , with e_k^c corresponding to $e_{F(k)}$, and to let

$$w_{jk} = \begin{cases} 1 & \text{if } j = F(k) \\ 0 & \text{if } j = F(k'), k' \neq k \\ a_{j,F(k)} / \sum_{k'} a_{j,F(k')} & \text{otherwise} \end{cases} \quad (4.8)$$

so that (4.7) is satisfied with

$$C_0 = \min_j \left(\sum_k |a_{j,F(k)}| / a_{jj} \right). \quad (4.9)$$

Hence, all we have to do to obtain good relaxability (large C_0) is to select the coarse level so that each fine-level variable j is strongly coupled with the coarse-level variables, i.e., so that the total of its coarse level couplings $(\sum_k |a_{j,F(k)}|)$ is a large fraction (C_0 , at least) of its total couplings $(a_{jj} = \sum_i |a_{ji}|)$.

This approach was indeed the first one adopted in AMG [B6]. It was later realized, however, that (4.7) can be satisfied, with about the same C_0 , by an interpolation I_c that uses fewer coarse-level points k for any fine-level point j (see Sec. 4.4). This is important in order to keep the coarse-level problem (4.2) as sparse as possible.

Observe that to satisfy (4.7) it is not necessary that large weights w_{jk} correspond to direct strong couplings $a_{j,F(k)}$. It is enough that they correspond to strong *connection* from j to $F(k)$ (cf. Sec. 4.1). If, for example, $a_{j,F(k)}=0$ but there exists an ℓ such that both $a_{j\ell}$ and $a_{\ell,F(k)}$ are strong, then we can use large w_{jk} , because we can satisfy (4.7) through

$$\begin{aligned} w_{jk}(e_j - e_k^c)^2 &= w_{jk}(e_j - e_\ell + e_\ell - e_{F(k)})^2 \\ &\leq \alpha a_{jk}(e_j - e_\ell)^2 + \beta w_{jk}(e_\ell - e_{F(k)})^2, \end{aligned} \quad (4.10)$$

for some $(\alpha-1)(\beta-1) = 1$. (The choice $\alpha=\beta=2$ may be standard, but one can take α/β to be an increasing function of $a_{j\ell}/a_{\ell,F(k)}$). Thus, each fine-level point j must have strong connections, but not necessarily direct couplings, to the coarse level.

The interpolation from strongly connected (not necessarily directly coupled) points is the usual practice in *geometric* multigrid, hence the

present theory applies to all such cases. It applies, in particular, to equations with strongly discontinuous coefficients, like those in [A1].

4.4 Strong coupling overlaps

Consider now the case where each fine-level variable j is strongly coupled to *many*, say roughly M , other variables; i.e., $O(M)$ terms should be taken in the summation of (4.9) to make $C_0 = O(1)$. This is likely to be the situation, for example, on coarse levels produced by the AMG procedure (4.2) (having not used in I_c the reduction in interpolation points being discussed here). Does this mean then that we must interpolate to each point from $O(M)$ points in order to obtain relaxability? It does, if the problem is arbitrary, but it does not, if it has (implicit) geometrical background. In the latter case, if each point has $O(M)$ strong neighbors, then those strong neighborhoods must have overlaps of size $O(M)$; i.e., each point j must have (many) neighbors i such that i and j have $O(M)$ common strong neighbors. Choosing even just one of these neighbors i to be a coarse-level point k , and using (4.10) for every λ which has strong couplings to both $i=F(k)$ and j , we get $O(M)$ strong-term contributions to majoring the left-hand side of (4.7), which is what we needed in order to satisfy (4.7) with $C_0 = O(1)$.

Thus, even for problems with large neighborhoods of strong couplings, we can still get good relaxability by interpolating to any point j from just few neighbors, provided those neighbors' neighborhoods largely overlap the neighborhood of j .

That such an overlap is *necessary* we can see from the following example.

Example 4.1. The fine grid is a rectangular MXN grid, so that each unknown i has a column number I_i and a row number J_i . The coefficients

of the matrix are defined by

$$a_{ij} = \begin{cases} M & \text{if } j=i \\ -1 & \text{if } j \neq i, I_j = I_i \\ -1 & \text{if } I_j = I_i - 1 \pmod{N}, J_j = J_i \\ 0 & \text{otherwise.} \end{cases}$$

It is easy to find (e.g., by mode analysis) that a point relaxation smoothes the error (very efficiently) along columns, but not along rows. Hence the coarse grid may represent even just one point per column, with interpolation within each column the convergence will be fast. On the other hand the convergence will be slow (convergence factor $1 - O(M^{-1})$) if we interpolate to each point from its left (mod N) neighbor, with whom the point has poorly overlapping neighborhoods.

4.5 Conclusions in more general cases

The above conclusions for the positive-type case can be extended to general locally positive definite matrices (sec. 2.1): By substituting $\sum_m f_m(e)^2$ for $e^T A e$ in (4.4), various interpolation strategies to guarantee relaxability, and various limitations, come into view. Basically, one can interpolate to a fine point j by taking that e_j which minimizes $\sum_m^j f_m(e)^2$, where $\sum_m^j$ is a sum over all those $f_m(e)$ which include e_j and which otherwise include only coarse-level variables $e_{F(k)}$. As in Sec. 4.4, this can then be modified to sums which include fewer terms. Practical ways of finding such interpolation when the f_m forms are not explicitly given are preliminarily discussed in [R1].

In case of *essentially positive type* matrices, for example, this approach shows that as long as the positive off-diagonal terms a_{ij} are small enough, the positive type procedures discussed above can remain unchanged. If, for example, for each $i \neq j$ with $a_{ij} > 0$ one can find

an ℓ such that $-a_{i\ell} > 4a_{ij}$ and $-a_{\ell j} > 4a_{ij}$, with each ℓ assigned to only one such (i,j) , then, by the argument of Sec. 2.5, condition (4.7) will retain its form, except that the sum $\sum_{j,i}$ would run only on the negative a_{ji} and some of these a_{ji} would be replaced by $a_{ji}/2$.

In more extreme cases of essentially positive type operators, however, the above approach shows that some modifications are required. Consider for example one of the most extreme cases of 7-point finite-difference approximation to second-order elliptic equations. This is the approximation to the (actually already parabolic) equation

$$U_{xx} - 2U_{xy} + U_{yy} = f,$$

based on the "wrong" seven points, yielding the difference stencil

$$\begin{pmatrix} 0 & 2 & -1 \\ 2 & -6 & 2 \\ -1 & 2 & 0 \end{pmatrix}. \quad (4.11)$$

In this case the positive form $E = e^T A e$ (which can be written in a local form by the procedure of Sec. 2.5, or otherwise) does no longer have terms of the form $b_{ij}(e_j - e_i)^2$. It is made solely of terms like

$$b_{\alpha,\beta} (2e_{\alpha,\beta} - e_{\alpha-1,\beta} - e_{\alpha,\beta+1})^2$$

or

$$b'_{\alpha,\beta} (2e_{\alpha,\beta} - e_{\alpha+1,\beta} - e_{\alpha,\beta-1})^2,$$

where (α,β) are the grid (column, row) numbers. This shows that in order to satisfy (4.4) one must use interpolation in which

$$w_{(\alpha,\beta)(\alpha+1,\beta)} = w_{(\alpha,\beta)(\alpha,\beta-1)} \text{ and } w_{(\alpha,\beta)(\alpha-1,\beta)} = w_{(\alpha,\beta)(\alpha,\beta+1)}. \quad (4.12)$$

Indeed, using interpolation which does not always satisfy (4.12) yielded in this case bad results [B6, Table 1].

In case of *block positive type* matrices the above approach tells us that we should interpolate along strong couplings in the sense anticipated in Sec. 3.5, i.e., interpolation along directions spanned by eigenvectors corresponding to large eigenvalues of $-A_{IJ}-A_{JI}$. These eigenvalues should not be small compared to couplings *within* the diagonal blocks. In case they are, and perhaps in any case of block positive type, the natural thing to do is to use the corresponding block relaxation. This requires a modification of our analysis, discussed next.

4.6 Two-grid analysis with block relaxation

In case of block relaxation, relaxability after the coarse-level correction is obtained if $\hat{R}^0 \geq C_0 E^0$, where $\hat{R}$ is defined in Theorem 3.5 and $\hat{R}^0$ is its value after the coarse-level correction. For this purpose we have the following generalization of Theorem 4.1.

THEOREM 4.2. *If for any fine-grid vector e there exists a coarse-level e^c such that*

$$C_0 \sum_J (e - I_c e^c)_J^T A_{JJ} (e - I_c e^c)_J \leq e^T A e \quad (4.13)$$

then $\hat{R}^0 \geq C_0 E^0$.

In (4.13) we used $(e - I_c e^c)_J$ to denote the J -th block of $e - I_c e^c$.

The proof is simply the block version of the proof of Theorem 4.1.

It is easy to see that this theorem implies relaxability for the usual anisotropic operators (such as $\varepsilon \partial_{xx} + \partial_{yy}$), provided the suitable block

relaxation (such as y-line relaxation). is employed. Theorem 4.1 shows, however, that block relaxation is not needed if interpolation has large weights only along strong connections (vertically, in the example).

Another case of block relaxation, sometimes called *collective relaxation*, is used when each block contains a fixed number q of unknown and stand for the values of q different functions at the same grid point. In this case the matrix is often *block positive type* or *essentially block positive type* (see Sec. 2). It is then natural to coarsen the problem using the same block structure: The coarse-level vector e^c is also made of q -blocks, its K -th block, denoted $\hat{e}_K^c$, usually corresponding to one of the fine-level blocks, denoted $\hat{e}_{F(K)}$. The *block interpolation* then have the form

$$\hat{e}_J = \sum_K W_{JK} \hat{e}_K^c, \quad (4.14)$$

where each W_{JK} is of course a $q \times q$ matrix. Theorem 4.2 then yields

COROLLARY 4.2. *If for any e there exists an e^c such that*

$$c_0 \sum_J (\hat{e}_J - \sum_K W_{JK} \hat{e}_K^c)^T A_{JJ} (\hat{e}_J - \sum_K W_{JK} \hat{e}_K^c) \leq e^T A e \quad (4.15)$$

then $\hat{R}^0 \geq c_0 E^0$.

In case of block positive type matrices one can use (2.7) to replace $e^T A e$ in (4.15) by

$$\frac{1}{4} \sum_{I,J} (\hat{e}_I - \hat{e}_J)^T (-A_{IJ} - A_{JI}) (\hat{e}_I - \hat{e}_J)$$

and derive block interpolation strategies analogous to those in Secs. 4.3 and 4.4.

4.7 Limitations of purely algebraic algorithms

In purely algebraic multigrid algorithm no use is made of any geometric information, including the geometric location of unknowns. (The geometric origin of the problem may still play an implicit role, as in Sec. 4.4.) This may lead to severe losses in convergence rates, as the following examples show. We later discuss how to mend such situations.

Example 4.2. We start with the 3-point one-dimensional Poisson equation, in which case

$$a_{ii} = 2, \quad a_{i,i+1} = a_{i,i-1} = -1.$$

The coarse grid points correspond to every other fine grid point, i.e., e_i^c represents e_{2i} . The only difference from the usual (geometric) multigrid treatment of this problem is in defining the interpolation weights: We assume interpolation from the right neighbor only:

$$w_{ik} = \begin{cases} 1 & \text{if } i=2K \text{ or } i=2K-1 \\ 0 & \text{otherwise.} \end{cases} \quad (4.16)$$

Suppose a smooth error e^- is given, with (slowly varying) slope s . Its coarse-grid approximation e^c , produced by (4.2), minimizes the L_2 norm of the slope of $e^- - I_c e^c$. But by (4.16) the slope of $I_c e^c$ vanishes on half the intervals, so all that can be minimized is the L_2 norm of the slope of $e^- - I_c e^c$ on the rest of the intervals. The minimum of the latter is actually zero, produced by e^c for which

$$e_k^c - e_{k-1}^c = \frac{1}{2} (e_{2K-1}^- - e_{2K-2}^-)$$

(except at the last interval). Thus, the slope of e^c is $s/2$. Hence the slope of $e^0 = e^- - I_c e^c$ is on the average $s - s/2 = s/2$, but it is actually

alternating between 0 and s . The following relaxation sweeps will efficiently reduce these oscillations (thereby reducing E from E^0 to $E^0/2$); but, no matter how many sweeps are made (if the grid is sufficiently large), the (average) slope will remain $s/2$. Thus the complete two-grid cycle can have a convergence factor no better than .5, no matter how many relaxation sweeps it employs.

Incidentally, the error e^0 produced in this example after the coarse-grid correction can serve as an example to an error e for which

$$E = \sum_{i,j} (-a_{ij})(e_i - e_j)^2 \ll \sum_i a_{ii} e_i^2,$$

but which nevertheless cannot be reduced by a coarse-grid correction (cf. Sec. 3.5).

The fault in the example above is not exactly the fact that we interpolate from one point only. To stress this point we bring now a modification of this example.

Example 4.3. Take now the usual 5-point two-dimensional Poisson equation, with the coarse grid comprising of the red points (grid points (α, β) such that $\alpha + \beta$ is even). Assume first that the interpolation to each black (i.e., not red) point employs its west and north neighbors only. By an argument similar to the above it can be shown that, ignoring some effects near boundaries, the complete two-grid cycle can reduce the north-west-to-south-east slope of the error at most by a factor .5, no matter how many relaxation sweeps are utilized.

If, on the other hand, the two neighbors used in interpolation are the west and east (or north and south) ones, we get quite a usual cycle, where convergence factors can be as small as we please, depending on using

enough relaxation sweeps (as can be verified by mode analysis, or by Sec. 5, or by other well-known theories).

Thus, geometry plays here a decisive role: Knowing in what direction to interpolate can dramatically change the convergence rates. This is even more so when multigrid V-cycles are used. Interpolations as in the above examples can be shown to give convergence factors no better than $1-2^{1-L}$ per cycle, where L is the number of levels.

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